2024 INFORMS Annual Meeting

Automating ML Workflow Orchestration
: Strategies for Autonomous Post-Deployment Model Updates

Adaptive Model Update Framework through dealing with Concept Drift

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Background

Modern Manufacturing Processes

- Key processes of modern manufacturing: **Continuous processes AI and data-driven Automation**
  - **Continuous Process**: Managing continuous data quality is a **challenge**. → **Changes over time.**
    - Essential to keep real-time **monitoring** and **response** across the process.
  - **AI and data-driven Automation:** Leverage to maximize efficiency. → **Why – When – How to Automize?**
    - **Machine Learning Operations(MLOps)** framework is crucial to automate Machine Learning (ML) model deployment, monitoring, and management.
- **Workflow Orchestration** in MLOps
  - Automates **data preparation** to model deployment through **feedback loops and monitoring components.**
  - Automates **retraining process** when a model performance **decreases** by continuous monitoring.
  - Automatically **updates models** to ensure accuracy in changing environments.
- Limitations of Workflow Orchestration
  - **Insufficient solutions for when and how to monitor the model's performance.**
  - **Lacks robust methods to address long-term changes in data and model performance.**

RQ1. **Why** do we have to automize the processes for monitoring and retraining?

RQ2. **When and How** to monitor and retrain for continuous training?

Background

Concept Drift (CD)

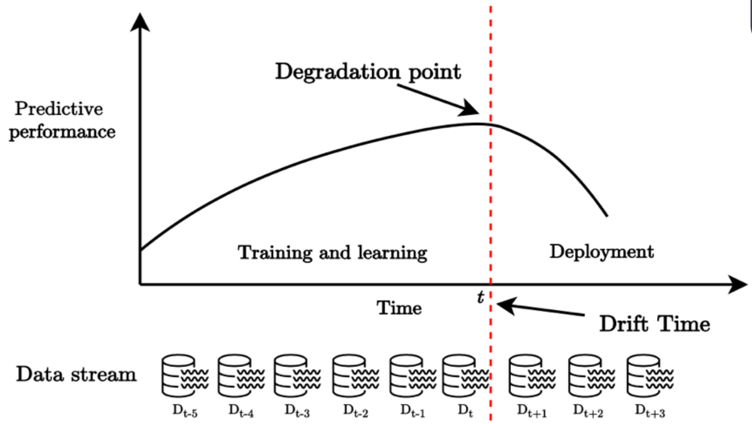

Figure 1. Concept Drift

$$P_{D_t}(X,Y) \neq P_{D_{t+1}}(X,Y)$$

Why? → **Concept Drift**

- The phenomenon where the statistical properties of the predictor variables and/or the target variable **change over time**.
    - Model quality depends on **detecting** and **responding** to concept drift effectively.
    - **Effective model management** requires proper detection and adaptation of CD.

- **Two Stages of CD Research: CD Detection and CD Adaptation**

When? → **Concept Drift Detection**

Measure changes and **determine the timing** for model update.

- Monitor the **performance metrics** as a **measurement** of CD.

How? → **Concept Drift Adaptation**

Management **methods for updating** the model effectively.

- Management of **Scheduler Design** and **ML Metadata Store**

- **Applying CD Detection and Adaptation methods to Workflow orchestration in MLOps** enables the automatic model updates, ensuring continuous training and monitoring of ML models.

Background

Limitations in Related Works

Table 1. Comparison of the proposed method with the existing CD detection and/or adaptation methods.

Study	Multiple windows and learners	Feature store updates	Data store updates	
			Buffer	Differential weighting
This study	○	○	○	○
PL	○	×	○	○
STEPD	○	×	×	○
fsNB	○	×	○	×
AL	○	×	×	×
STUDD	○	×	×	×
MWDDM-H/M	×	×	○	○
DDM	×	×	○	×
EDDM	×	×	○	×
RDDM	×	×	○	×
MDDM-A/G/E	×	×	○	×
FHDDM	×	×	○	×
HDDM-A/W	×	×	○	×
FW-DDM	×	×	○	×
BDDM	×	×	○	×
ADWIN	×	×	×	×
ECDD	×	×	×	×
EDIST1 & 2	×	×	×	×
SeqDrift1 & 2	×	×	×	×

○: Addressed, ×: Not Addressed

1 Only Periodic Retraining

- Risks missing ML model's performance degradation points.

2 Simple Retraining for Degradation

- Ignores shifts in feature importance from environmental changes.

3 Neglect Updating Feature & Data Stores

- Hinders adaptation to evolving data environments.

Proposed Method

Adaptive Model Update Framework

1 Non-Periodic Retraining

- Utilizes multiple windows to detect performance degradation dynamically.

2 Pattern-Based Retraining Strategies

- Considers shifts in feature importance and diverse change scenarios.

3 Feature & Data Store Updates

- Incorporates feature selection and dynamic window management with buffer periods and differential weighting.

Workflow Orchestration Components

Scheduler Design	• Helps model to enhance performance and adapting it to changing data patterns. • **Dynamic Detection**: Using multiple memory models to detect CD and classifies it into 3 types. • **Feature selection**: Select relevant features for model updates, ensuring optimal model accuracy.
ML Metadata Store	• Concerns how to structure the data for retraining the ML model. • Essential to focus on the quantity and importance of data in model update. • **Buffer Periods** with **Differential Weighting** are applied to adjust the model update.

Proposed Method

Adaptive Model Update Framework

Scheduler Design with CD Detection Methods

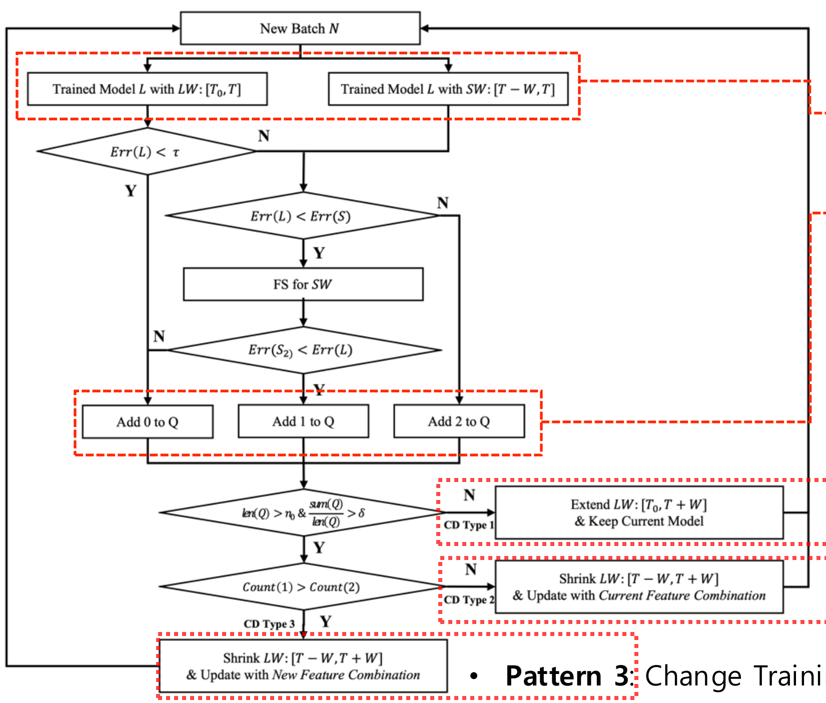

Figure 2. Adaptive Model Update Framework

Pattern-based CD Detection with Multiple Models

- **Multi learners:** Monitors **long-term (LW) and short-term (SW) memory models** to detect concept drift and trigger updates.
- **Measurement for CD Detection**:
  - **Queue management system (Q)** monitors the model's performance over time. (value of 0, 1, 2)
  - τ, δ, and $maxq$ **:** Hyperparameters for controlling system's sensitivity - How **quickly detect** data change.
- **Pattern 1**: Extend Training Window (LW+) & Maintain Existing Model
- **Pattern 2**: Change Training Window LW → SW & Maintain Existing Model
- **Pattern 3**: Change Training Window LW → SW & **Select New Optimal Feature Combinations**

- Improves model accuracy and stability with appropriate retraining strategy.

Proposed Method

Adaptive Model Update Framework

ML Metadata Store Update with CD Adaptation

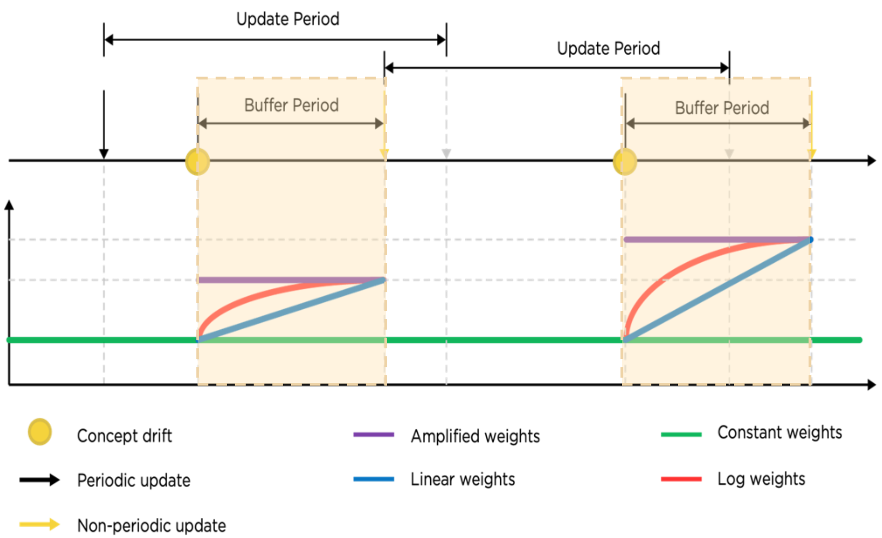

Figure 3. Adaptive Strategy for Process Changes Using Differential Weighting and Buffer Period

Buffer Period

- A period that collects additional data **before retraining after detecting CD.**
- **Shorter** buffer period: **Quicker** adaptation
- **Longer** buffer period: **Slower** adaptation

Differential Weighting

- A weighting scheme that **applies varying importance** to buffer period data during model updates.
- **Adjust sensitivity** to new distributions during model updates
- Constant, Amplified, Linear, Log Weights

Case Study

Model Update Problem in Steel Industry

- **Continuous Process in Steel Manufacturing**
  - **Sequential Stages**: Ironmaking → Steelmaking → Casting → Rolling
  - **Return Fines Monitoring**: Predicts sinter quality. Sinter quality affects final steel product and its quality.
- **Operational Challenges**
  - Data pattern changes over time, but there is no effective update strategy.
  - Hard to identify degradation causes without stopping production.
  - **Need for workflow orchestration** to ensure stability and performance.
- **Performance Degradation of Return Fines Predictive Model**

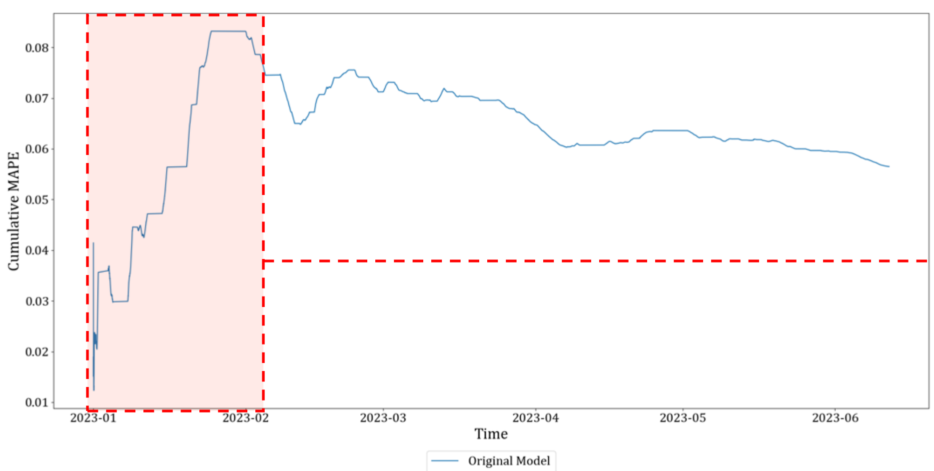

Figure 4. Cumulative MAPE and performance decline over time

- **Virtual Metrology System Implementation**
  : ML-based, 30 variables, predicts sinter yield, and performance metric CTQ(%) and MAPE.
- **Performance Degradation Problem (CTQ)**
  : 65% (2021) → 51.86% (Jan-Jun 2023)
- Implemented the adaptive model update strategy to maintain the performance of the predictive model.

Case Study

Application of Adaptive Model Update Framework

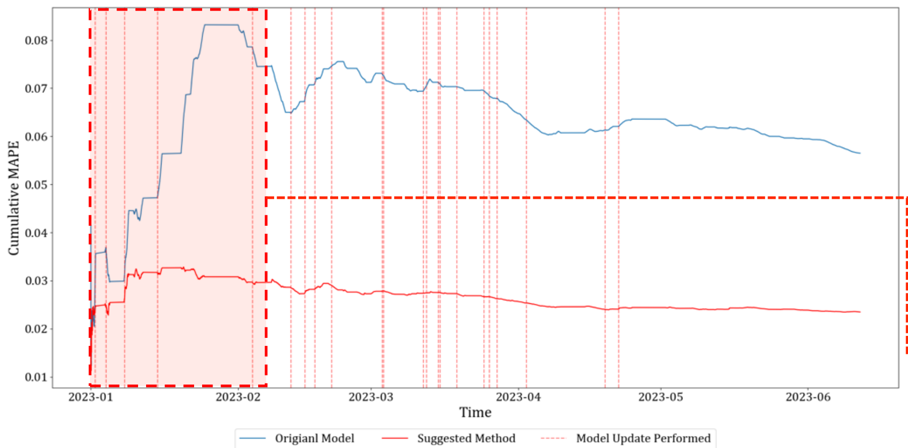

Figure 5. Comparative cumulative MAPE of models over time

Table 2. Comparison of the performance of various models

	Adaptive Update		
	Buffer Period	**Weight**	**Performance**
Original	X	X	**5.65 / 53.06**
Proposed	X	Constant	2.35 / 92.39
		Double	2.35 / 92.41
	O	Constant	2.36 / 92.29
		Double	**2.34 / 92.47**

Performance Metrics : MAPE / CTQ (%)

Performance Comparison (MAPE / CTQ (%))

- Overall Performance
  - **Original** : 5.65 / 53.06
  - **Proposed**: 2.34 / 92.47
- Performance in January ~ February 2023
  - **Original** : 6.56 / 48.83
  - **Proposed**: 2.79 / 91.40

Key Insights

- Reduces performance drops and boosts accuracy.
- Applied immediate updates at model update periods to prevent further decline.
- The model adapts to changes, balancing drift sensitivity and stability.

Discussion

Impact of Pattern-based CD Detection

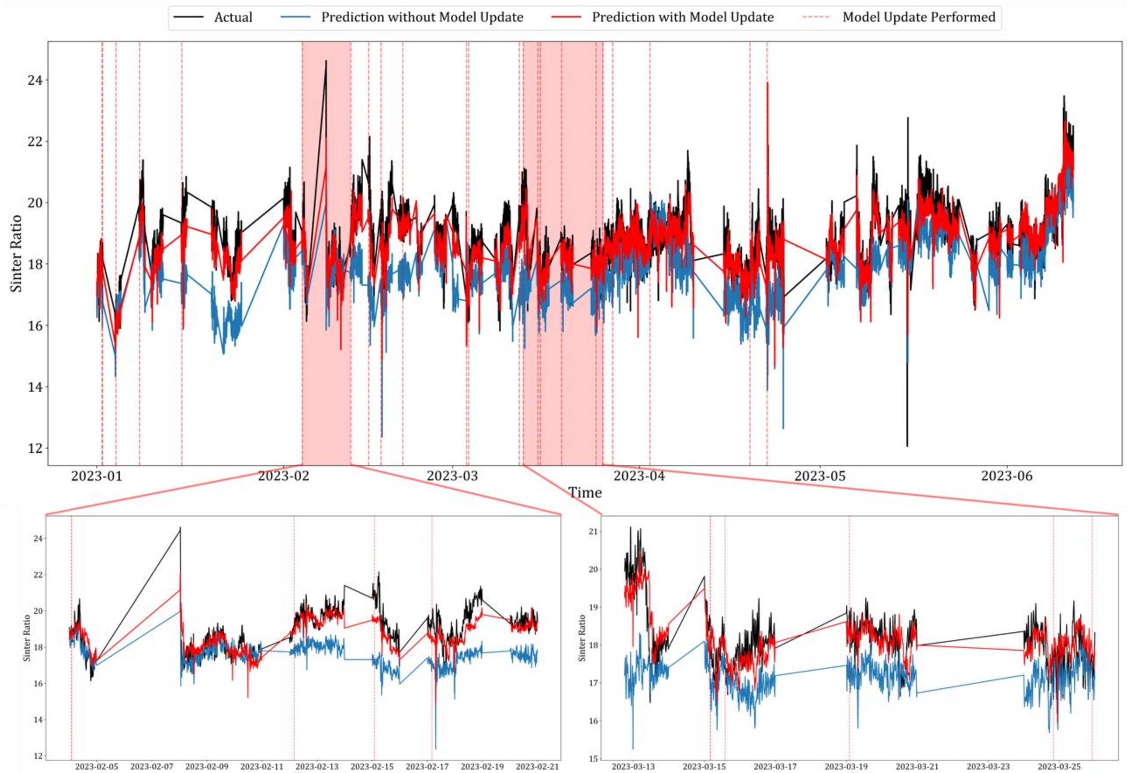

Figure 6. Comparison of Model Predictions over time

Performance Comparison (MAPE / CTQ (%))

- Overall) 2023-01-01 ~ 2023-06-11

  Original: 5.65 / 53.06

  Proposed: 2.34 / 92.47

- Period 1) 2023-02-04 ~ 2023-02-12

  Original: 7.04 / 43.67

  Proposed : 2.78 / 89.43

- Period 2) 2023-03-09 ~ 2023-03-26

  Original: 6.87 / 31.94

  Proposed : 2.56 / 88.10

- The performance stabilized after applying the model update strategy.

Key Insights

- Detect and adapt to sudden drops and rebounds in model performance.
- Maintain stable predictive performance during abnormal periods.

Discission

Impact of Buffer Period and Differential Weights for CD Adaptation

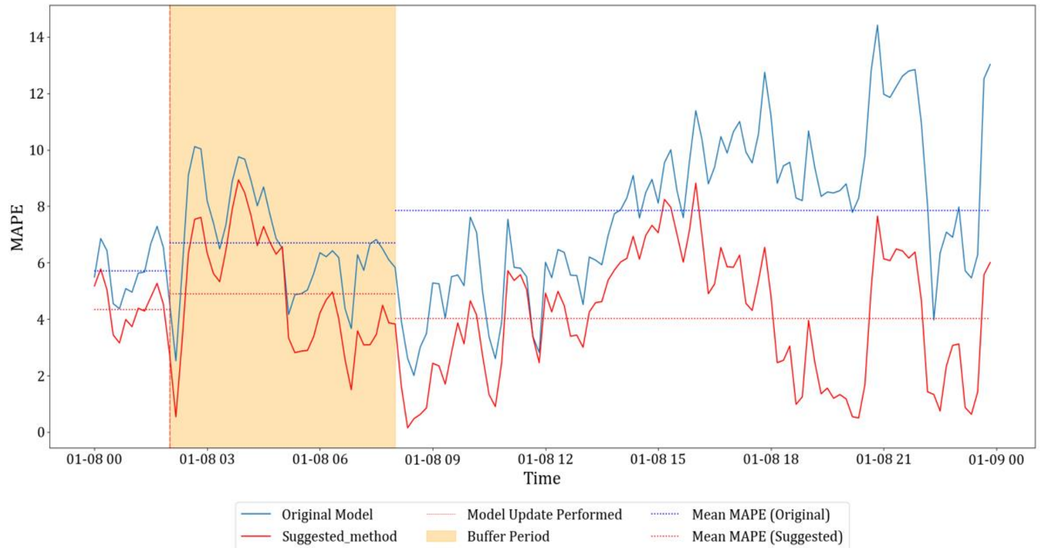

Figure 7. Impact of the buffer period on cumulative MAPE for models

Table 3. Impact of Buffer Period and Differential Weights

Period	Original	Proposed
Overall	7.39 / 7.63	4.29 / 40.97
Before Buffer	**6.70 / 0**	**4.88 / 30.76**
Buffer	5.70 / 5.40	4.33 / 37.83
After Buffer	**7.84 / 9.37**	**4.03 / 43.75**

Performance Metrics : MAPE / CTQ (%)

Impact of Buffer Period and Differential Weighting (MAPE / CTQ (%))

- Original: Higher and greater variability
  - Before Applying Buffer: 6.70 / 0
  - After Applying Buffer: 7.84 / 9.37
- **Proposed**: **Lower** and **stabilized** performance
  - Before Applying Buffer: 4.88 / 30.76
  - After Applying Buffer: 4.03 / 43.75

Key Insights

- Mitigated performance degradation during sudden data shifts.
- Ensured stability and accuracy in real-time predictive environments.

Conclusion

Summary – AS-IS vs. TO-BE

1 Only Periodic Retraining

- Risks missing ML model's performance degradation points.

1 Non-Periodic Retraining

- Utilizes multiple windows to detect performance degradation dynamically.

2 Simple Retraining for Degradation

- Ignores shifts in feature importance from environmental changes.

2 Pattern-Based Retraining Strategies

- Considers shifts in feature importance and diverse change scenarios.

3 Neglect Updating Feature & Data Stores

- Hinders adaptation to evolving data environments.

3 Feature & Data Store Updates

- Incorporates feature selection and dynamic window management with buffer periods and differential weighting.

Conclusion

Key Findings

1 **Non-periodic retraining demonstrates both high adaptability and stability when updating the model, while also preventing further performance decline.**

2 **Pattern-Based Retraining Strategies are effective to maintain stable predictive performance, especially during abnormal periods.**

3 **Feature and data store updates based on buffer periods and differential weighting schemes are effective in ensuring stability and accuracy in real-time predictive environments.**

Future Works - Dynamic Retraining Interval Strategies

- Investigate strategies for adaptive retraining intervals based on data volatility.
- Explore real-time feedback mechanisms for optimizing retraining cycles.

Thank you for listening!